Student Performance Predictor

Submitted by:

Malaika Bala

Course:

Fundamentals of AI and ML

Institution:

VIT Bhopal University

Date:

23 March 2026

---

1. Introduction

The Student Performance Predictor is a simple project developed to estimate a student's final marks based on key academic factors such as study hours, attendance, and previous performance. The purpose of this project is to demonstrate how basic concepts of artificial intelligence and machine learning can be applied to solve real-world problems in education.

---

2. Problem Statement

Students often do not have a clear understanding of how their study habits and consistency affect their academic performance. There is a need for a simple system that can provide an estimate of their final marks based on measurable inputs.

---

3. Objective

- To predict a student's final marks
- To analyze the effect of study hours, attendance, and previous marks
- To demonstrate the application of basic AI/ML concepts
- To build a simple and user-friendly program

---

4. Methodology

The project is developed using a simple predictive approach. The program takes three inputs from the user:

- Study hours
- Attendance percentage

- Previous marks

A formula is applied to these inputs to calculate the predicted final marks. The weights assigned to each factor represent their importance in determining performance.

---

5. Tools and Technologies Used

- Python (Programming Language)
- Basic programming concepts (input/output, conditions, calculations)

---

6. Working of the Project

1. The user runs the program
2. The program asks for input values
3. The inputs are processed using a formula
4. The predicted marks are calculated
5. The result is displayed to the user

---

7. Results

Sample Input:

- Study Hours: 5
- Attendance: 80%
- Previous Marks: 70

Output:

Predicted Final Marks: 81.0

The program successfully predicts marks based on the given inputs.

---

8. Challenges Faced

- Designing a simple prediction logic
- Handling user inputs correctly
- Ensuring the output remains within valid range

---

9. Conclusion

This project demonstrates how basic concepts of AI and logical problem-solving can be used to predict student performance. It provides useful insights into how different factors influence academic results and helps students improve their study habits.

---

10. Future Scope

- Use real machine learning algorithms
- Train the model on a larger dataset
- Develop a graphical user interface (GUI)
- Convert into a web or mobile application